



Energy Efficient Clustering with Optimized Action Sequences based Q-Learning Algorithm for Routing in Heterogeneous WSN

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Abstract: Wireless Sensor Networks (WSNs) consist of low-power sensor nodes that communicate with a central hub to collect and transmit data. Since these nodes operate on limited battery power, energy efficiency is a critical concern. Traditional routing protocols often struggle with dynamic topology changes, heterogeneous device support, and scalability limitations. To address these challenges, this paper proposes an Energy-Efficient Clustering with Optimized Action Sequences-Based Q-Learning Algorithm for routing in heterogeneous WSNs. The proposed approach leverages Q-learning to determine optimal routing paths by selecting Cluster Heads (CHs) based on attribute-based clustering, while neighboring nodes form clusters. The CHs aggregate data from cluster members and forward it to the sink node. Additionally, the Adaptive Coati Optimization Algorithm (ACOA) enhances the action sequences of Q-learning, preventing isolated nodes and further optimizing routing efficiency. The proposed Q-Learning-based Routing Algorithm (QLRA) achieves optimal routing by dynamically adjusting Q-values based on selected action sequences. The performance of the approach is evaluated through extensive simulations conducted in MATLAB under three distinct scenarios: varying the number of rounds, the number of nodes, and the percentage of CH selection. The proposed Optimized QLRA with clustering and ACOA are compared against three baseline methods, such as Optimized QLRA without clustering, Traditional QLRA with clustering, and Traditional QLRA without clustering. The results demonstrate a significant 11.82% increase in throughput and a 25.19% reduction in energy consumption, confirming the proposed algorithm's effectiveness in improving energy efficiency, network lifetime, and data delivery reliability.

Keywords: Sensor nodes, Sink, Cluster head, Optimal routing, Q-Learning, ACOA, Forwarder nodes, WSN.

1. Introduction

WSNs have gained significant attention across various sectors due to their ability to support diverse applications such as production, healthcare, public transportation, education, agriculture, architecture, energy management, commerce, and banking [1]. In recent years, wireless sensor networks have been increasingly recognized as a fundamental component of modern technological systems, undergoing rapid evolution [2]. As part of the everywhere computing initiative, wireless sensor networks are considered to be a prime component. A vast number of tiny, low-power sensors collaborate to gather and send data wirelessly over great distances in wireless sensor

networks. In the last couple of decades, with the advancement of wireless and network technologies, a wide variety of applications have been developed, such as tracking, environmental monitoring, etc. [3, 4].

WSNs typically use a lot of energy because, in order to operate effectively, they must interface with other nodes and provide data regularly [5]. In WSNs, energy efficiency is an important consideration, since they need to be able to maintain their operational capabilities for long periods of time, in order to be successful [6]. A sensor node is a battery-operated device, so maximizing the life of the network is one of the most important design parameters to consider when designing the node [7]. WSNs can be broadly classified into different types based on their network

architecture and node capabilities. These include homogeneous WSNs, where all sensor nodes have identical hardware and energy constraints, and heterogeneous WSNs, where nodes possess varying energy levels, computational power, and communication ranges [8]. Heterogeneous WSNs are widely used in large-scale deployments as they improve network scalability, data aggregation efficiency, and energy optimization [9-11]. In heterogeneous WSNs, where nodes differ in energy capacity, computational power, and communication range, energy efficiency strategies must account for these variations. Unlike homogeneous WSNs, where all nodes operate under similar constraints, heterogeneous WSNs leverage high-energy sensor nodes that handle most communication and processing tasks, thereby reducing the burden on low-energy nodes. This hierarchical structure helps optimize energy consumption, enhance data aggregation, and improve network scalability [12]. In most cases, these nodes are deployed in remote or harsh environments, making it difficult for them to be properly replaced or recharged regularly. Energy usage is dependent on the number of sensor nodes deployed in addition to the nodes' distance from one another, frequency of data collection, and number of deployments. It is crucial to carefully take into account each of these elements in order to make the network more energy efficient and prolong its battery life [13-15].

Therefore, maintaining network uptime and prolonging its lifespan depend heavily on energy efficiency. Additionally, the type of communication protocol used and the data processing techniques employed can also impact energy consumption. By selecting energy-efficient routing protocols and implementing data compression algorithms, we can minimize the energy requirements of their sensor networks while still obtaining accurate and timely data. By optimizing energy consumption, WSNs can achieve longer network lifetime, improved scalability, and reduced. While traditional routing protocols have been effective in optimizing certain aspects of WSN, they often fall short when it comes to balancing energy consumption across the network. Many studies have introduced energy-saving techniques, but these often face limitations such as high computational overhead, increased communication costs, or insufficient scalability. Our method stands out by addressing these challenges through a novel combination of clustering algorithms and an optimized Q-learning-based routing mechanism. Clustering reduces energy consumption by limiting the number of nodes that need to communicate directly with the base station, thereby reducing

overall transmission energy. The Q-learning-based routing mechanism dynamically adjusts the routing paths based on real-time network conditions, ensuring that the most energy-efficient routes are selected. This adaptability is key in prolonging network lifespan, as it distributes energy usage more evenly across the network.

Among the main contributions of the paper are:

- ❖ To improve energy efficiency, heterogeneous nodes in the network are clustered in this approach. At first, cluster heads are selected using an attribute-based CH selection method.
- ❖ Moreover, data from cluster members is gathered and transmitted to the SINK via CH. For that, CH installs the optimal routing path to the SINK.
- ❖ For optimal routing, an Optimized Action Sequences Based Q-Learning Algorithm is presented in this approach.
- ❖ In order to prevent isolated nodes from existing within the network, the next forwarder nodes or actions of the Q-learning model are selected optimally using adaptive coati optimization algorithm.

The following is a broad outline of the paper. In Section 2, we present the results of the various studies on improving the energy efficiency of Wireless Sensor Networks that have been conducted by various researchers. Our study methodology is outlined in Section 3. As a result of several strategies, Section 4 describes their consequences and results, while Section 5 summarizes their findings and offers recommendations for future research.

2. Literature survey

Rowayda A. Sadek et al [16] introduced an energy-efficient routing protocol based on the grey wolf optimizer (GWO) with the Tabu search algorithm (TSA). GWO was incorporated into the CH selection process as part of the GWO-TSA procedure. The best path from the CH to the BS was chosen by TSA employing forwarding of reliable route packets and a fitness function that depends on the energy residual and distance of the sensor node from the base station. In addition, TSA discovered multiple paths from CH to BS that are optimum for data transfer in an effort to increase data transmission dependability. The number of sensor nodes had little effect on GWO-TSA's effectiveness; nevertheless, the dependability of the network and the node's energy & distance from the base station had a significant impact.

Bhukya Sureshand and G. Shyama Chandra Prasad [17] introduced an energy-efficient safe

routing system in WSNs for IoT networks based on the LEACH protocol. LEACH was a cluster-based protocol that divided the network using clusters. Using the random numbers provided by every sensor node, LEACH selected a subset of nodes to act as cluster chiefs. A trust-based secure relay selection technique was introduced to the routing path in order to improve choosing of relay nodes and secure data transfer even further. However, the drawbacks of this technique include high computation complexity, time consuming, and difficulty in tuning the parameters.

An energy-efficient routing protocol for heterogeneous wireless sensor networks called CREEP was presented by SunitiDutt et al [18]. Through modification of the CH selection thresholds inside a heterogeneous two-level WSN; this method used the CREEP protocol to increase network longevity. With BS at its center, CREEP views a disc of radius that approximates the square WSN field as a circular one. Any CH lying on this disc would have sent its aggregated data in a single hop to the BS. The CHs that were farther away from the BS than the radius distance sent their aggregated data to a CH that was nearby instead of directly to the BS. Additionally, as cluster communication may be burdened by this strategy, network reliability and latency may be impacted.

Arun Kumar Rana and Sharad Sharma [19] presented a hybrid directing convention for IoT situations: Heterogeneous WSN-capable Advanced Zone-Stable Election Protocol (AZ-SEP). In this protocol, certain nodes sent data to the base station lawfully, while other nodes sent data to the base station by bunching. The AZ-SEP was put into practice and contrasted with the conventional Low Energy adaptive clustering hierarchy (LEACH). The recreation's findings demonstrated that, in comparison to earlier conventions like LEACH, AZ-SEP improved both the even time frame with throughput. Although the proposed AZ-SEP convention outperforms the existing LEACH convention, it cannot rapidly choose the right cluster head during data transmission.

An energy-efficient distance-based clustering protocol for heterogeneous WSN was presented by M. Sheik Dawood et al. [20]. With this technology, energy-efficient communication throughout the dispersed nodes in the field was made possible by CH deployment between the nodes. This strategy divides the region into four equal halves, known as the Quad Zone. In a zone, they chose a Cluster head that was close to the zone's center, so that all member nodes in that zone could communicate effectively. Even though the wireless sensor network's lifespan is

extended by this energy-efficient routing protocol, it does not provide effective routing between nodes.

A heterogeneous ring clustering method and E2HRC routing protocol were developed by Guangjie Han et al. [21] to balance the used energy and lower overall energy usage. They created a backbone network routing scheme according to hop difference, node residual energy, and optimal direction angle. This method outperformed the original IPv6 routing system in terms of overall efficiency and average energy usage, all while maintaining equilibrium across the wireless sensor network. E2HRC, on the other hand, has a greater packet loss ratio.

The Energy-Efficient Cooperative Routing Scheme for Heterogeneous WSN (EERH) was developed by K.L. Tsai et al. [22]. By deploying WSNs in an identical geographic area and relaying packets produced by both the sensor network and other WSNs, this strategy creates heterogeneous sensor networks. Moreover, to save delivery energy and inform sinks more effectively, packets that a sensor directed in the same direction were combined into a single packet. Simulation results have demonstrated that the EERH extends a heterogeneous WSN's lifetime efficiently. However, it has the possibility that a packet may be lost if the source and the sink are in the same direction.

Hye-Jin Kim et al. introduced Energy Centers Searching using Particle Swarm Optimization (EC-PSO), a novel clustering method [23]. In this procedure, the CHs were first selected by a geometric process. Clustering was done using EC-PSO after the network's energy started to vary. Energy centers were identified using an improved PSO technique, and nodes close to the energy center were selected to act as CHs. A protective mechanism to prevent low-energy nodes from acting as the forwarder was also implemented, along with a mobile data collector to gather the data. However, packet loss during transmission could result from the mobile data collector's erratic behavior.

Moreover, Bairagi, Dutta, and Babulal [24] introduced the Geographic Forwarding Energy Efficient Routing Protocol (GF-EERP), which enhanced the Geographic Energy Aware Routing (GEAR) protocol by incorporating node categorization, a unique region head selection mechanism, multi-hop communication, and the removal of dead nodes. Similarly, Sabri and Hilmani [25] presented the ProFor and EnProFor protocols, which relied on a predefined relay probability for successful message delivery to the base station, emphasizing the analysis of relaying nodes and relay probabilities to ensure robust performance amid

individual node failures. In another approach, Dogra, Rani, and Gianini [26] developed an improved region-based routing protocol (REERP) for IoT WSNs that enhanced network longevity through methods such as the addition of new nodes to existing clusters and energy-aware head node selection, outperforming traditional protocols in terms of lifetime and energy consumption. Lastly, Thangavelu and Rajendran [27] proposed the Energy-Efficient Memetic Clustering Method (EEMCM), which merged the Parallelized Memetic Algorithm (PMA) with the AlexNet architecture to enhance anomaly detection efficiency in IoT.

The existing studies in energy-efficient routing protocols exhibit several limitations that can impact their performance and adaptability. For instance, the GF-EERP relies heavily on accurate node categorization, which can falter in dynamic conditions, and its single head node selection may not effectively adapt to fluctuating energy levels. Similarly, protocols like ProFor and EnProFor depend on predefined relay probabilities, limiting their adaptability to sudden topology changes. The REERP faces stability issues when integrating new nodes into clusters, which can lead to energy holes due to uneven energy distribution. Additionally, while the EEMCM leverages complex algorithms for anomaly detection, it may struggle with real-time performance due to computational overhead. To address these limitations, the proposed methodology integrates an optimized Q-learning-based routing algorithm that enhances the energy efficiency and performance of WSN. The use of Q-learning facilitates real-time optimization of routing decisions, allowing the network to dynamically respond to changes in energy levels and node status. Consequently, this innovative approach not only promotes equitable energy distribution but also improves the resilience of the network against node failures, ultimately leading to enhanced performance across varied operating conditions.

3. Proposed methodology

3.1 System model

Fig. 1 shows an illustration of the proposed system model. The system model includes sensor nodes (SN), which are all heterogeneous nodes, the Cluster Head, and the SINK. In the system, the sensor nodes are linked wirelessly, dispersed in clusters with a CH, and the data is collected, transmitted, and processed by the system. As part of the data collection process, CHs collect data from SNs, data is

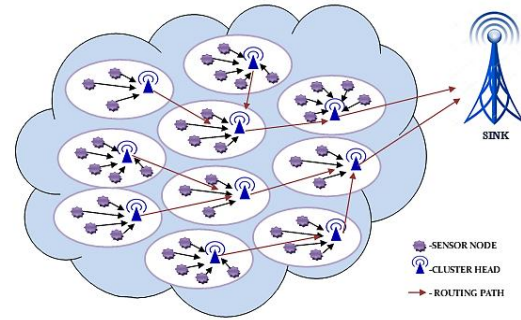


Figure. 1 Proposed System Model

Table 1. Notations used in this study

Notation	Description
SN	Sensor nodes
CH	Cluster Head
FN	Forwarder Node
L_{QF}	Link quality factor
P_N	Neighborhood proximity value
RE	Residual Energy of a node
C_{SN}	Current node
N_{SN}	Neighbouring node
$dist(C_{SN}, N_{SN})$	The distance between node n and the neighboring node
$H_r(s)$	Hop Count between the sender and the SINK.
$E_r(s)$	sender's current energy
$Q_r(s)$	sender's Q-value
α	Learning rate controls how much new information affects the Q-value.
γ	The discount factor determines the importance of future rewards
ϵ	Exploration probability in ϵ -greedy policy

transmitted to the SINK by optimal routing, and the SINK processes and analyzes the data. Table 1 depicts the notations used in this study.

3.1.1. Brief overview of methodology

Intending to improve the energy efficiency of heterogeneous WSNs, we propose a method based on clustering with a Q-learning-based routing algorithm. For cluster head selection, cluster heads are selected using the attribute-based CH selection method. After selecting CH, nearby nodes in its communication range are joined as cluster members. For optimal routing, a Q-Learning-based Routing Algorithm is presented in this approach as it collects data from cluster members and forwards it to the SINK via the optimal routing path. To avoid the existence of isolated nodes in the network, next forwarder nodes or actions of the Q-learning model are selected optimally using Adaptive Coati Optimization

Algorithm. Based on the selected action sequences, the Q-value is generated for each node in the network. Depending on the Q-value, optimal routing is attained. Fig. 2 depicts the overall structure for the proposed methodology.

3.2 Attribute-based CH selection method

In a WSN, the heterogeneous sensor nodes are first clustered together to increase network efficiency. Using an attribute-based technique, cluster heads (CHs) are chosen for clustering. To improve the suggested approach's optimality and network performance, choosing a suitable CH is essential. In order to choose the best CH, the CH selection method considers variables including the communication link quality factor (LQF), remaining energy of SNs, and neighborhood proximity value. Below is a detailed explanation of the selection process:

The neighbor node's proximity is indicated in the following manner:

$$P_N = \frac{1}{N_{total}} \sum_{j=1}^{N_{total}-1} dist(n, j) \quad (1)$$

Whereas $dist(n, j)$ is the distance measure between node n and its neighbor node j , and N_{total} denotes the total number of neighbor nodes in the wireless sensor network.

As follows, the L_{QF} among the source and destination nodes is mentioned:

$$L_{QF}(s, d) = \sum_{j=0}^n (S_j + R_j + B_j + I_j) \quad (2)$$

Where, S_j represents the Queue size, R_j represents the Response time, B_j represents the Bandwidth of the link, I_j and represents the interference ratio.

S_j is mathematically defined as,

$$S_j = \frac{TotalS_j - OcupiedS_j}{Framelength} \quad (3)$$

The following is a description of the response time estimation:

$$R_i = (T_{DRP} - T_{DTP}) + (T_{ATP} - T_{ARP}) \quad (4)$$

Where the duration of the data packet enqueue at the receiving port is defined by T_{DRP} . Time spent de-

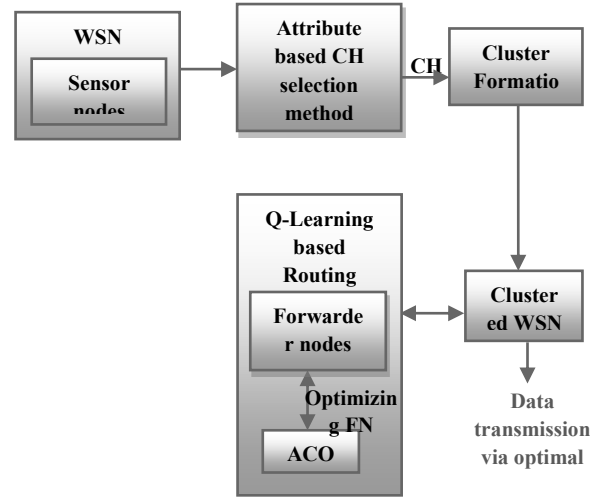


Figure. 2 Overall structure for proposed methodology

queuing data packets at transmitting ports is defined by T_{DTP} ; time spent en-queuing acknowledgement packets at transmitting ports is defined by T_{ATP} ; and time spent de-queuing acknowledgement packets at receiving ports is defined by T_{ARP} .

To select the CH, sensor nodes can consider factors such as NP, LQF between source and destination nodes, remaining energy of sensor nodes. Once the CH is selected, nearby nodes within its communication range can join as cluster members. The CH can then distribute energy resources among cluster members to ensure balanced energy consumption and prolonged network lifetime. By applying these, sensor nodes can efficiently select cluster heads based on attributes such as link quality, energy levels, and proximity, thus optimizing the performance of the wireless sensor network.

3.3 Q-Learning-based routing algorithm

Following clustering, the data is gathered by the CH and sent to the SINK by the cluster members. The CH transmits the data directly to the SINK if it is only one hop away. Otherwise, the CH installs the optimal routing path to the SINK. For optimal routing, QLRA is implemented in this approach. In QLRA, a variant of Q-learning, every node keeps a Q-table that contains the expected reward for each possible action (i.e., forwarding the packet to a neighbouring node) in each possible state (i.e., the current location of the packet in the network). Each sensor node in QLRA is treated as an agent that chooses the subsequent forwarding node based on its Q-value, an assessment of the total reward transmitted by the current node to the SINK via the subsequent node. Q-value is updated based on the reward the agent receives after performing an action. To ensure effective learning and decision-making in the proposed QLRA-based

routing protocol, several core parameters are defined within the Q-learning framework. An exploration–exploitation strategy is adopted using an ϵ -greedy approach, allowing sensor nodes to balance between exploring new forwarding paths and exploiting known optimal routes. Initially, nodes are encouraged to explore the network, while over time, the strategy shifts toward exploiting the best-learned actions based on accumulated experience. The learning rate (α) governs how strongly new information influences existing Q-values, while the discount factor determines the importance of future rewards in evaluating current actions. Together, these parameters enable each node to learn energy-efficient and stable routing behaviors over time. The termination of an episode is defined by the successful delivery of a packet to the SINK node or by reaching a predefined maximum hop count, ensuring that learning episodes remain bounded and avoid routing loops. These elements collectively contribute to the adaptability and convergence of the QLRA model under dynamic WSN conditions.

The agent chooses the course of action that maximizes the predicted cumulative reward after gradually finding out what every decision's true Q-value is. In order to determine the optimal pathways and maximize the network lifetime, the QLRA protocol considers variables including residual energy, connection distance, and hop count.

The next forwarding node in the proposed QLRA protocol is chosen based on Q-values, which are also utilized to modify the Q-value of the current node. In the subsequent data transmission phase, the preceding node might utilize the feedback contained in the data packet to modify its Q-value. The network gets configured first. Every node transmits a control packet to each of its neighbouring nodes during this phase, beginning at the SINK. The control packet also contains the location of the sender, remaining energy, along the hop count to the SINK. After acquiring a control packet, the receiver node (k) uses Eq. (8) to determine the sender's Q-value $Q_r(s)$ by extracting the sender's information. Next, the sender's details are entered into the receiving SN (k) neighbour table. Receiving SN (k) calculates its hop count towards SINK using Eq. (9), and transmits its own data with the control packet.

$$Q_r(s) = \frac{E_r(s)}{H_r(s)} \quad (8)$$

$$H_r(j) = H_r(s) + 1 \quad (9)$$

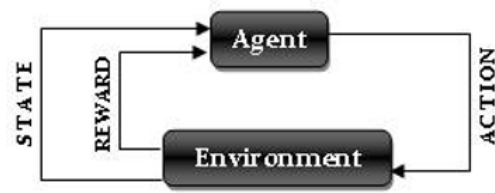


Figure 3 Basic Q-learning approach

Packet ID
Node ID
Residual Energy
Location coordinate
Hop Count
Q-Value
Next Forwarder
Payload Data

Figure 4 Structure of Data Packet

Here $H_r(j)$ and $H_r(s)$ stand for the number of hops from the sender and node j , respectively, to the SINK.

Fig. 4 depicts the data packet's structure. Every sensor node refreshes its neighbor database and grasps information from packets it receives or overhears. It drops the packet when the current node does not match the sensor node that a Next Forwarder field indicates. If not, packet forwarding responsibility will be taken on by the current node. As a result, the current node searches for neighboring nodes first before forwarding the packet. If the SINK node is in the neighbor table, the packet is delivered straight to the SINK. If not, this node searches its neighbor table for potential forwarder nodes.

It needs to fulfil the following requirements in order to be considered as a potential neighbour node:

- The candidate neighbour node has a Q-value of not zero, which indicates that it does not have a node that forwards to the SINK.
- From the neighbor node, it is fewer hops to the SINK than from the current node.
- Compared to the current neighbour node, the candidate neighbour node is closer to the SINK.

No neighbor node that meets the aforementioned requirements turns an isolated node and sets its Q-value to 0. Then, the current node uses Eq. (10) to determine each possible neighbour node's pertinent Q-value if there are several of them.

$$Q_{new}(C_{SN}, N_{SN}) = (1 - \alpha)Q_{old}(C_{SN}, N_{SN}) + \alpha(R(C_{SN}, N_{SN}) + Q(N_{SN})) \quad (10)$$

The learning rate is represented by α , then the estimated path quality via a certain neighbor node from the present node to the SINK is indicated by $Q(C_{SN}, N_{SN})$ the reward for the current node for sending a packet to a particular neighbor node is represented by $R(C_{SN}, N_{SN})$. Path quality from this neighbor node to the sink is indicated by $Q(N_{SN})$. The neighbor table provides $Q(N_{SN})$, and Eq. (11) can be used to compute $R(C_{SN}, N_{SN})$.

$$R(C_{SN}, N_{SN}) = \frac{E_r(N_{SN})}{dist^n(C_{SN}, N_{SN})} \times H_r(N_{SN}) \quad (11)$$

where $E_r(N_{SN})$ denotes the neighbour node's residual energy, then $H_r(N_{SN})$ denotes the neighbour node's hop count to the SINK. Those two are available from the neighbor's table. The distance, $dist(C_{SN}, N_{SN})$, among the current node as well as its neighbor could be calculated using Eq. (12). Furthermore, the value of the constant n is displayed in Eq. (13).

$$dist(C_{SN}, N_{SN}) = \sqrt{(x(N_{SN}) - x(C_{SN}))^2 + (y(N_{SN}) - y(C_{SN}))^2} \quad (12)$$

Here $(x(C_{SN}), y(C_{SN}))$ denotes the current node's location coordinates, $(x(N_{SN}), y(N_{SN}))$ and denotes the neighbour node's location coordinates.

From Eq. (11), the reward $R(C_{SN}, N_{SN})$ in the QLRA protocol is determined by three factors. These factors are

- The neighbour node's residual energy,
- The hop count of the neighbour node to the SINK, and
- Distance from the current node and its neighbour.

The reward is inversely correlated with the neighbor node's hop count to SINK and proportional to the neighbor node's residual energy. By doing this action, packet delivery might be enhanced, and node energy use can be balanced within the network. Furthermore, the reward $R(C_{SN}, N_{SN})$ is inversely related to the fourth power of the distance if it exists a distance higher than the threshold $dist_0$ among the current node as well as the neighbor node. In the absence of this, the reward is inversely related to the square of the distance.

Eq. (10) is utilized by the current node to determine which potential neighbor node with the highest $Q(C_{SN}, N_{SN})$ is the next forwarder. Eqs. (14) and (15) are then utilized to update the node's original Q-value and hop count.

$$Q(C_{SN}) = \max_{N_{SN} \in N} Q(C_{SN}, N_{SN}) \quad (14)$$

where (N_{SN}) indicates any one of these nodes, N indicates the set of possible neighbor nodes of the current node, and $Q(C_{SN})$ denotes the current node's Q-value. In other words, the greatest $Q(C_{SN}, N_{SN})$ represents the strength of the best connection between the current node and the SINK.

$$H_r(C_{SN}) = H_r(N_{SN}) + 1 \quad (15)$$

Here (N_{SN}) represents the node selected as next forwarder, then $H_r(C_{SN})$ as well as $H_r(N_{SN})$ denotes the hop count of the current node and the chosen node, respectively.

Next, the packet is sent to the following forwarder by the current node, which modifies information contained in the packet header (position coordinate, node ID, hop count to the SINK, residual energy, and Q-value). This packet is observable to the preceding node as well as the Q-value within it could be interpreted as feedback from the present node to the preceding node. This feedback for the preceding node could be applied to the subsequent data transmission cycle. Energy can be saved by taking the action of encoding feedback into the data packet.

3.3.1. Optimizing action sequences using ACOA

ACOA is used to select optimally the next forwarder nodes or actions of the Q-learning model in order to prevent the occurrence of isolated nodes. By using ACOA, the Q-learning model can select forwarder nodes with high residual energy and minimum distance from the sink, which will reduce the communication and computation time of the system. This is due to the fact that nodes with high residual energy and the closest distance to the sink will be able to offer faster transmission times and higher communication quality. When this ACOA is implemented, the system's total calculation and communication times will be shortened. Moreover, ACOA can also help to prevent the occurrence of isolated nodes, which could lead to inefficient routing.

ACOA is a novel meta-heuristics algorithm inspired by coatis' behaviour in nature. It mimics their packing and hunting tactics, especially targeting green iguanas. Coatis, known for their omnivorous diet, often forage for iguanas collectively, with some members scaring them towards the ground while others pack them. However, early convergence issues in the COA algorithm led to local optima traps. To address this, Levy Flight, known for diversifying search agents to avoid local minima, is integrated

with COA. This integration enhances optimization efficacy by preventing premature convergence.

Here are the steps involved in ACOA:

I. Initialization

The population, represented as the Y matrix, has been initialized by coati. Coati's positions can be represented by Y . Here, forwarder nodes ($FN_1, FN_2, \dots, FN_N$) in the network are considered as solutions in this study. Here is the format of the initial solution

$$Y_i = \{y_1, y_2, \dots, y_n\} \quad (16)$$

In this case, we have Y_i representing the population of the solution, n representing the variables, and then,

$$y_1 = \{FN_1, FN_2, \dots, FN_N\} \quad (17)$$

The example solution set is depicted in Table 2.

Consider the solution set of forwarder nodes, which is represented in binary form; here, 1 represents "TRUE" and 0 represents "FALSE". If FN is "1", that node has to be included for optimal routing, or else if it is "0" means that node should be neglected. This process is repeated for every iteration to find the optimal solution set.

II. Fitness function

The fitness of each search agent is then evaluated for each iteration using conventional benchmark methods. Therefore, this study project uses an optimized search to identify the best set of forwarder nodes that may generate the highest fitness score, which has maximum residual energy as well as minimum distance from the sink. As part of the ACOA, each coati's fitness is evaluated using the following formula:

$$fitness = \max \sum \left(TotalRE_{FN} + \frac{1}{dist(C_{SN}, BS)} \right) \quad (18)$$

Here $TotalRE_{FN}$ is the Total residual energy of the selected forwarder nodes, and $dist(C_{SN}, BS)$ it is the distance between the Current node and the sink.

III. Updation phase

The ACOA update was inspired by coatis hunting and attacking iguanas in their habitat and fleeing predators in their habitat. A candidate solution is updated based on this behavior by the ACOA.

Table 2. Example solution set

Solution	FN	FN	FN	FN	...	...	FN
s	1	2	3	4			N
Solution Set 1	1	0	0	1	...	...	1
Solution Set 2	0	1	1	1			0
Solution Set 3	1	1	0	0	...	...	0
:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:
:	:	:	:	:	:	:	:
Solution Set N	0	0	1	0	...	...	1

Phase 1: Hunting and attacking strategy on the iguana

Taking into account that the iguana is placed at the best position in the population, half of the coatis ascend up trees while the other half waits on the ground for it to fall. The following formula can be used to simulate the coatis' position on the tree:

$$Y_i^{p1}: y_{i,j}^{p1} = y_{i,j} + rand \cdot (IGU_j - I \cdot y_{i,j}) \quad (19)$$

In order to find the location of a fallen spiny-tailed iguana on the floor, we use the equation as follows:

$$IGU_j^l = LB_j + rand \cdot (UB_j - B_j) \quad (20)$$

$$Y_i^{p1} = \begin{cases} y_{i,j} + rand \cdot (IGU_j^l - I \cdot y_{i,j}), & f_{IGU^l} < f_i \\ y_{i,j} + rand \cdot (y_{i,j} - IGU_j^l), & \text{else} \end{cases} \quad (21)$$

$$Y_i = \begin{cases} Y_i^{p1} & f_i^{p1} < f_i \\ Y_i & \text{else} \end{cases} \quad (22)$$

In this section Y_i^{p1} indicates the new location estimated for the i th coati; $y_{i,j}$ signifies the j -th dimension of the coatis; and f_i^{p1} defines the value of the objective function. $rand$ Represents a random real number in the range $[0,1]$, and IGU denotes wherever an iguana is in the search space, although it truly refers to where the best member is, IGU_j represents the j -th dimension number, and I designates an integer randomly chosen from the set $\{1, 2\}$. IGU^l Indicates the iguana's randomly generated location on the ground, whereas $iguana_j^k$ indicates the j -th dimension of the iguana's position. f_{IGU^l} Provides the objective function value, whereas $\lfloor \cdot \rfloor$ indicates the floor function.

The introduction of the Levy flight mechanism during the exploration stage improves the flexibility and convergence speed of the algorithm, thus significantly enhancing its search capability. The formula is presented below:

$$\vec{Y}_{(it+1)} = \vec{Y}_{it} + \mu \text{sign} \left[\text{rand} - \frac{1}{2} \right] \oplus \text{levy} \quad (23)$$

Herein, $\vec{Y}_{it}$ denotes the i^{th} coati's location at iteration it , μ represents a uniformly distributed random value, $\oplus$ denotes the mean entry-wise product, and then rand denotes a random number ranging within $[0, 1]$. Then the $\text{sign}[\text{rand} - 1/2]$ only had three values (1, 0, -1) accordingly. The Levy Flight created the random walk distributions shown below.

$$\text{levy} \sim u = t^{-\lambda}, 1 < \lambda \leq 3 \quad (24)$$

Levy flight step durations are as follows:

$$s_l = \frac{\mu}{|v|^{1/\beta}} \quad (25)$$

λ is constructed using the formula for $\lambda = 1 + \beta$ where $\beta = 1.5$ and $\mu = N(0, \sigma_\mu^2)$ we have similar regular stochastic distributions with

$$\sigma_\mu = \left[\frac{\frac{\Gamma(1+\beta) \sin(\pi\beta)}{2}}{\Gamma\left(\left(1 + \frac{\beta}{2}\right) \right) \beta \Gamma\left(\frac{\beta-1}{2}\right)} \right] \text{ and } \sigma_v = 1 \quad (26)$$

Phase 2: The process of escaping from predators

Coatis possess a survival instinct that drives them to run from predators. For the ACOA, such conduct directs the optimization process.

$$Y_i^{p2}: y_{i,j}^{p2} = y_{i,j} + (1 - 2\text{rand}) \cdot (LB_j^{\text{local}} + \text{rand} \cdot (UB_j^{\text{local}} - LB_j^{\text{local}})) \quad (27)$$

$$Y_i = \begin{cases} Y_i^{p2} & f_i^{p2} < f_i \\ Y_i & \text{else} \end{cases} \quad (28)$$

In this section f_i^{p2} stands for the coati's objective function value, $y_{i,j}^{p2}$ represents the coati's j -th dimension, Y_i^{p2} represents the updated location of the i th coati, rand as a random real integer in the

interval $[0, 1]$, and t for the iteration number. UB_j^{local} and LB_j^{local} , respectively, denote the local upper as well as lower limits of the j -th decision variable, and UB_j and LB_j , respectively, the upper as well as lower bounds of the j -th decision variable.

IV. Termination

The procedure of updating the population continues till the last iteration of the algorithm. The result of the ACOA run is the optimal solution that excludes isolated nodes and consists of a collection of forwarder nodes that reflect the best routing between the current node and SINK.

Algorithm: Pseudo-code of ACOA

Input: Forwarder nodes

Output: Optimized forwarder nodes

Start ACOA

Set the total number of iteration st , the number of coatis n , and the set of forwarder nodes $FN_1, FN_2, \dots, FN_N$.

Initialize the position of coatis and calculate the objective function

For $T = 1: t$

Update the iguana's location based on where the optimal member of the population is located.

Exploration phase: Attacking behavior

For $i = 1: \lceil n/2 \rceil$

Calculate the updated location for the i^{th} coati using Eq. (19)

Update the i^{th} coati's location using Eq. (22)

End For

For $i = 1 + \lceil n/2 \rceil: n$

Calculate a random location for the iguana using Eq. (20)

Calculate the updated location for the i^{th} coati using Eq. (21)

Update the i^{th} coati's location using Eq. (22)

Change the exploration agent's search location using Levy Flight (23)

End For

Exploitation phase: Escaping behavior

Calculate the local bound for variables using Eq. (27).

For $i = 1: n$

Calculate the updated location for the i^{th} coati using Eq. (28)

Update the i^{th} coati's location using Eq. (29)

End For

Save the position and save the fitness value of the best-fitted population.

End For

Return the Optimal solution by ACOA for the given problem.

End ACOA

It is therefore the ACOA that returns the optimized forwarder nodes or actions resulting from the Q-learning model, which may later be used in order to determine the optimal routing path. For each optimized Forwarder node, a Q-value is calculated. The Q-value is used to determine the importance of the optimized Forwarder node in the optimal routing path. The higher the Q-value, the more important the optimized Forwarder node is. This helps to identify the best routing path. On the basis of those Q-values, an optimal routing path is determined, and the CH (current node) transmits the data packets to the SINK over the optimized Forwarder nodes in the optimal routing path. For example, if Forwarder nodes (FN_1 , FN_4 , FN_9 , FN_{11} , and FN_{25}) are selected for the optimal routing path, the Q-value of FN_1 will be highest, and it will act as the optimized Forwarder node for the optimal routing path. Here, CH_0 will send the data packets from CH_1 which is the next forwarder node FN_1 in the optimal routing path. CH_1 will then forward the data packets to CH_4 , which is FN_4 , and so on until the SINK is reached. The optimized forwarding path is shown in Fig. 6 below.

4. Results and discussions

This section discusses the efficacy of the proposed methodology based on evaluation metrics and performance results compared to other relevant works. The proposed model is tested using a computer running Windows 10 with an Intel Core i7 processor, 8GB of RAM, and the Python programming language. The MATLAB Platform is used to carry out the proposed method. Here, we show the results for two distinct scenarios, such as round variability and node variability. These specifics are covered in the sections that follow. Table 3 depicts the simulation settings.

Scenario 1: Throughput performance with different CH selection strategies

We are considering two parameters, m and b , which play a crucial role in determining the performance of WSN. The parameter m represents

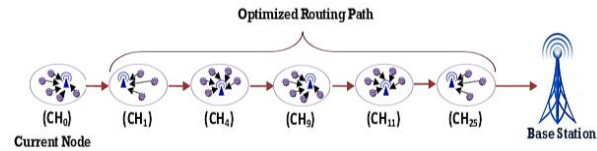


Figure. 6 Optimal Routing path example

Table 3. Simulation settings

Parameter	Value
Number of Nodes	300 nodes
Routing protocol	Optimized QLRA
Packet size	512bits
MAC	802.11
Radio propagation model	Two Ray Ground
Antenna	Omni Antenna
Initial Energy	0.5 J
Simulation time	100secs
Rate	500kb
Episode	1000
Discount factor	0.99
Learning rate	0.001
Initial Epsilon	1
Final Epsilon	0.1
Total number of coatis	30
Population Size	30
Upper bounds	99
Lower bounds	0
Maximum number of iterations	150
Number of runs	20

the fraction of advanced nodes, which possess higher energy than normal nodes, while b denotes the fraction of intermediate nodes with energy levels between normal and advanced nodes. Increasing m adds more advanced nodes to the network, enhancing overall energy availability and improving data transmission efficiency. Similarly, a higher b value increases the number of intermediate nodes, which also contributes to network performance but with a lesser impact compared to m .

Figs. 7 and 8 illustrate the relationship between throughput (in Kbps) and the number of rounds, considering different CH selection limits. The performance of the system is analyzed for five CH selection strategies: no limit, 30% CH, 20% CH, 10% CH, and 5% CH. The results indicate that a higher number of CHs leads to increased throughput, as more CHs facilitate efficient packet transmission to the BS. The no-limit CH selection provides the highest throughput, whereas the 5% CH selection exhibits the lowest throughput due to a reduced number of CHs, leading to data congestion and increased packet loss.

Fig. 7 represents the scenario with $m = 0.2$ and $b = 0.3$, while Fig. 8 considers $m = 0.3$ and $b = 0.3$. Increasing m from 0.2 to 0.3 results in higher throughput since advanced nodes have higher energy levels, contributing to better packet transmission. Since b (intermediate nodes) remains constant at 0.3, the primary improvement is driven by the presence of additional advanced nodes. The findings confirm that while increasing the number of CHs enhances throughput, an optimal balance must be maintained to ensure a prolonged network lifetime.

Scenario 2: Performance analysis based on round variability

In this scenario, we compare the proposed Optimized QLRA (with clustering) with existing methods such as Optimized QLRA (without clustering), Traditional QLRA (with clustering), and Traditional QLRA (without clustering) in terms of Delivery Ratio, Network Life Time, Throughput, Delay, Energy Consumption, and Total Residual Energy.

Fig. 9 above represents a comparative analysis of the Delivery Ratio for the proposed Optimized QLRA (with clustering) and existing methods, including Optimized QLRA (without clustering), QLRA (with clustering), and QLRA (without clustering), across different network conditions. Notably, the "Proposed Optimized QLRA-with clustering" consistently achieves the highest Delivery Ratios, starting at 0.848 for 100 rounds and steadily increasing to an impressive 0.997 for 300 rounds. The "Optimized QLRA - without clustering" method achieved a better Delivery Ratio as the number of rounds increases, reaching 0.981. Conversely, "QLRA - with clustering" and "QLRA - without clustering" methods exhibit lower but relatively stable Delivery Ratios, with improvements as round variability expands, reaching values between 0.685 to 0.968 and 0.735 to 0.9, respectively. These data demonstrated how well the "Proposed Optimized QLRA-with clustering" performs in delivering packets consistently across a range of round variability, making it the most effective solution for high and constant delivery ratios.

Fig. 10 compares throughput for Proposed Optimized QLRA (with clustering) with existing methods, such as QLRA (with clustering), optimized QLRA (without clustering), and QLRA (without clustering). With the "Proposed Optimized QLRA - with clustering", we consistently achieve the highest throughput values, ranging from 53 to 70 Kbps. Meanwhile, with the "Optimized QLRA - without

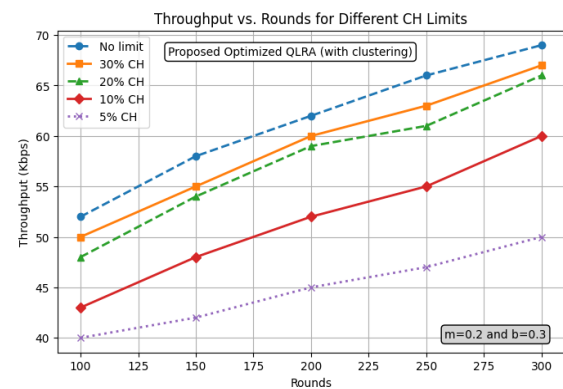


Figure. 7 Throughput Performance with Different CH Selection Strategies for $m=0.2$ and $b=0.3$

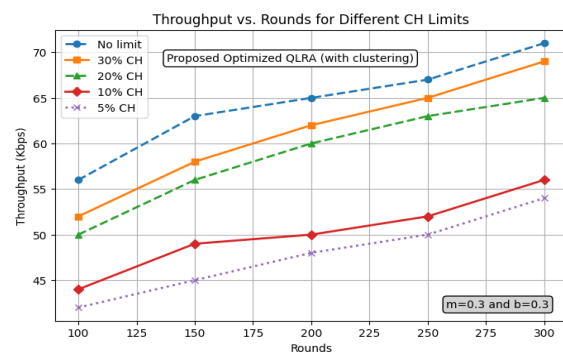


Figure. 8 Throughput Performance with Different CH Selection Strategies for $m=0.3$ and $b=0.3$

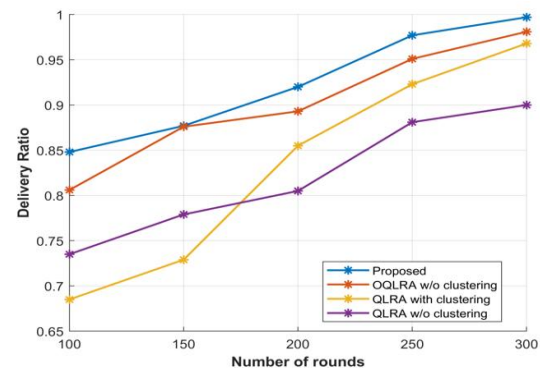


Figure. 9 Comparative Analysis based on Delivery ratio

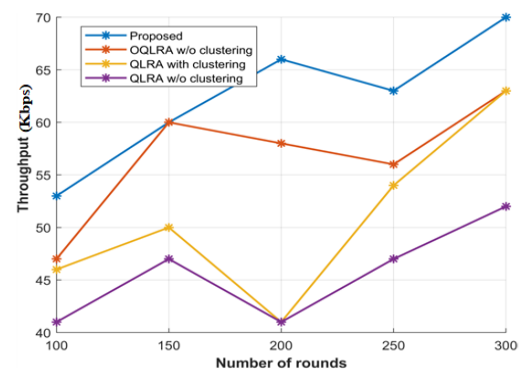


Figure. 10 Comparative Analysis based on Throughput

clustering", achieve throughput values of 47 to 63 Kbps. In contrast, the "QLRA-with clustering" and "QLRA-without clustering" methods exhibit variable throughput, with values 63 and 52 Kbps, respectively. Based on these measurements, our proposed "Proposed Optimized QLRA - with clustering" achieved superior throughput in all rounds when compared to existing methods, demonstrating high performance in all rounds.

Fig. 11 represents a comparative analysis of Network Lifetime for several routing methods, including the proposed Optimized QLRA (with clustering) and existing methods such as Optimized QLRA (without clustering), QLRA (with clustering), and QLRA (without clustering). There is a consistently stable network longevity range of 5.331 to 8.205 units for the "Proposed Optimized QLRA - with clustering". The "Optimized QLRA - without clustering" method shows Network Lifetime values ranging from 4.923 to 7.6277. As opposed to this, QLRA methods with and without clustering exhibit comparatively shorter Network Lifetimes, which are 4.122 and 6.324 units, respectively. From these results, the 'Proposed Optimized QLRA-with clustering' has significant advantages in terms of network resilience and network lifespan over different rounds of optimization.

As illustrated in Fig. 12, the proposed and existing methods are analyzed based on Network Lifetime. Our "Proposed Optimized QLRA - with clustering" consumes less energy, with a range of 44 J to 57 J for rounds 100 to 300, than existing methods. According to the "Optimized QLRA - without clustering" method, 58 J to 77 J of energy are consumed. As opposed to this, QLRA methods with or without clustering exhibit comparatively high energy consumption which ranges from 73 J to 90 J and 85 J to 99 J respectively. Based on these results, it is evident that the 'Proposed Optimized QLRA - with clustering' consumes less energy over different iterations. This indicates that the proposed and existing methods consume less energy when clustering is implemented.

Fig. 13 illustrates the comparison of the proposed methods and existing methods based on delay. Compared to existing methods, our "Proposed Optimized QLRA-with clustering" has a minimum delay of 1.454s to 4.12s. Based on the "Optimized QLRA - without clustering" method, 1.496s to 6.322s of delay was achieved. The delay for QLRA methods with and without clustering is comparatively high, ranging from 4.92s to 7.622s and from 5.328s to 8.2s, respectively. The results indicate that the 'Proposed Optimized QLRA-with clustering' consumes less

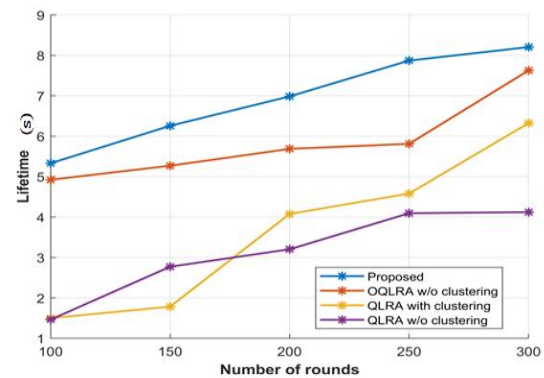


Figure. 11 Comparative Analysis based on Network Lifetime

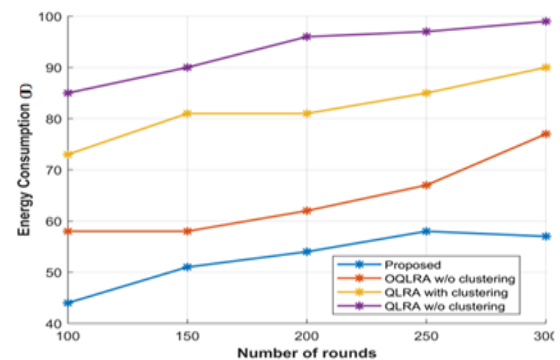


Figure. 12 Comparative Analysis based on Energy Consumption

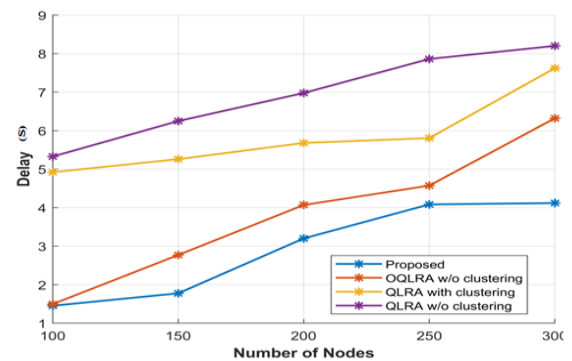


Figure. 13 Comparative Analysis based on Delay

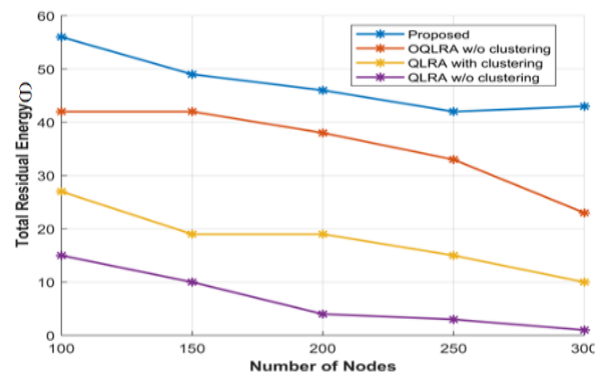


Figure. 14 Comparative Analysis based on Total Residual Energy

delay over multiple iterations. The clustering technique proved to be effective in reducing the overall delay of the method.

Fig. 14 presents a comparative analysis based on Total Residual Energy for proposed and existing methods. Total Residual Energy is a critical metric that reflects the remaining energy in all network nodes, providing insights into the overall energy sustainability. Notably, the "Proposed Optimized QLRA-with clustering" maintains a higher Total Residual Energy, ranging from 56 to 43 units across rounds 100 to 300. In contrast, the "Optimized QLRA - without clustering" method achieves Total Residual Energy values ranging from 42 to 23 units. In comparison, both QLRA methods, with and without clustering, exhibit lower Total Residual Energy levels, ranging from 27 to 10 units and 15 to 1 unit, respectively. These results underscore the "Proposed Optimized QLRA-with clustering's" ability to preserve more significant Total Residual Energy compared to the existing methods, indicating its superior energy sustainability.

Complexity analysis

Fig. 15 illustrates the variation in time per round for different routing methods as the number of rounds increases. It is evident that the proposed Optimized QLRA consistently outperforms the other methods in terms of computational efficiency. For example, at 300 rounds, the proposed method records a time per round of only 2.2s, whereas QLRA without clustering reaches 7.0s, indicating a substantial reduction in execution time. This improvement is attributed to the use of attribute-based clustering and optimized forwarder selection, which collectively reduce redundant route exploration and local decision overhead. The steady and minimal growth in execution time also confirms that the proposed approach scales efficiently with increasing rounds.

Scenario 3: Performance analysis based on node variability

This scenario 3 compares the Optimized QLRA (with clustering) with existing methods based on node variability, including Optimized QLRA (without clustering), Traditional QLRA (with clustering), and Traditional QLRA (without clustering) from the perspective of Delay, Delivery Ratio, Network Life Time, Throughput, Energy Consumption, and Total Residual Energy.

Fig. 16 above represents a comparative analysis of the Delivery Ratio for the proposed Optimized QLRA (with clustering) and existing methods,

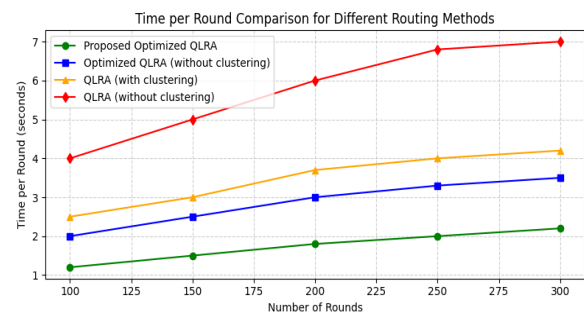


Figure. 15 Time per round comparison for different routing methods

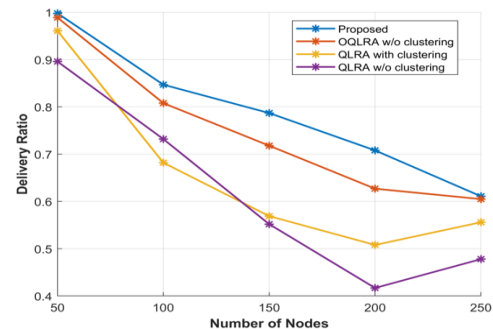


Figure. 16 Comparative Analysis based on Delivery Ratio

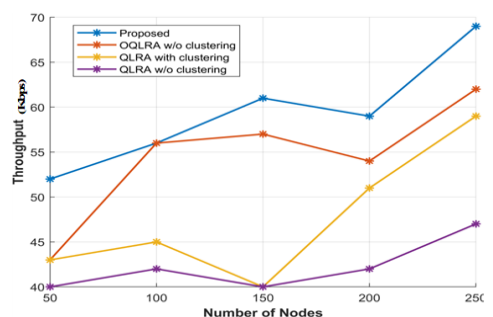


Figure. 17 Comparative Analysis based on Throughput

including Optimized QLRA (without clustering), QLRA (with clustering), and QLRA (without clustering), across different network conditions. Notably, the "Proposed Optimized QLRA-with clustering" consistently achieves the highest Delivery Ratios, starting at 0.998 for 50 nodes and steadily increasing to an impressive 0.611 for 250 nodes. The "Optimized QLRA - without clustering" method achieved a better Delivery Ratio as the number of nodes increases, reaching 0.989 to 0.605. Conversely, "QLRA-with clustering" and "QLRA-without clustering" methods exhibit lower but relatively stable Delivery Ratios, with improvements as node variability expands, reaching values between 0.961 to 0.556 and 0.896 to 0.478, respectively. These data demonstrated how well the "Proposed Optimized QLRA-with clustering" performs in delivering packets consistently across a range of node variability, making it the most effective solution for high and constant delivery ratios.

Fig. 17 compares throughput for Proposed Optimized QLRA (with clustering) with existing methods, such as QLRA (with clustering), QLRA (without clustering), and Optimized QLRA (without clustering). With the "Proposed Optimized QLRA - with clustering", we consistently achieve the highest throughput values, ranging from 52 Kbps to 69Kbps across 50 to 250 nodes. Meanwhile, with the "Optimized QLRA - without clustering", achieve throughput values of 43Kbps to 62Kbps. In contrast, the "QLRA-with clustering" and "QLRA-without clustering" methods exhibit variable throughput, with values from 43 Kbps to 59 Kbps and 40 Kbps to 47 Kbps, respectively. Based on these measurements, our proposed "Proposed Optimized QLRA - with clustering" achieved superior throughput in all nodes when compared to existing methods, demonstrating high performance in all nodes.

Fig. 18 represents a comparative analysis of Network Lifetime for several routing methods, including the proposed Optimized QLRA (with clustering) and existing methods such as Optimized QLRA (without clustering), QLRA (with clustering), and QLRA (without clustering). There is a consistently stable network longevity range of 5.002s to 7.593s for the "Proposed Optimized QLRA - with clustering". The "Optimized QLRA - without clustering" method shows Network Lifetime values ranging from 3.987s to 6.656s. As opposed to this, QLRA methods with and without clustering exhibit comparatively shorter Network Lifetimes, which are from 1.059s to 5.596s and 1.137s to 4.001s, respectively. It is clear from these results that the 'Proposed Optimized QLRA-with clustering' has significant advantages in terms of network resilience and network lifespan over different nodes of optimization.

As illustrated in Fig. 19, the proposed and existing methods are analyzed based on Network Lifetime. Our "Proposed Optimized QLRA - with clustering" consumes less energy, with a range of 50J to 36J for nodes 50 to 250, than existing methods. According to the "Optimized QLRA - without clustering" method, 71 J to 52 J of energy is consumed. As opposed to this, QLRA methods with or without clustering exhibit comparatively high energy consumption, which ranges from 69 J to 84 J and 79J to 93 J, respectively. Based on these results, it is evident that the 'Proposed Optimized QLRA-with clustering' consumes less energy over different iterations. This indicates that the proposed and existing methods consume less energy when clustering is implemented.

Fig. 20 illustrates the comparison of the proposed

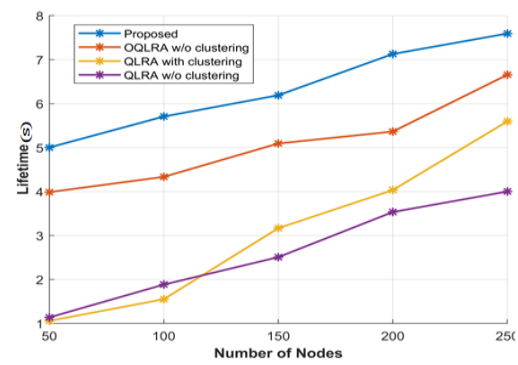


Figure. 18 Comparative Analysis based on Network Lifetime

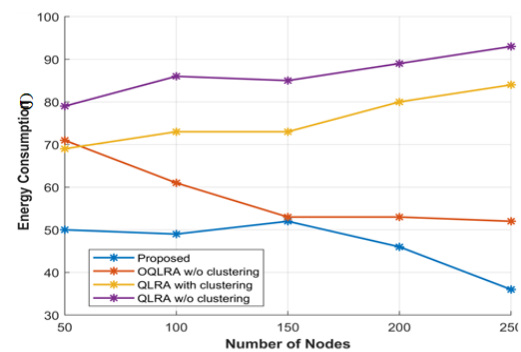


Figure. 19 Comparative Analysis based on Energy Consumption

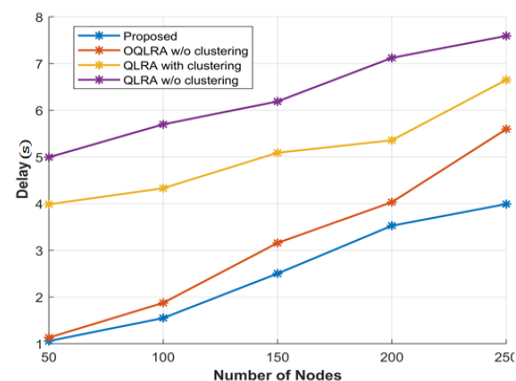


Figure. 20 Comparative Analysis based on Delay

methods and existing methods based on delay. Compared to existing methods, our "Proposed Optimized QLRA-with clustering" has a minimum delay of 1.058s to 3.991s. Based on the "Optimized QLRA - without clustering" method, 1.131s to 5.595s of delay were achieved. The delay for QLRA methods with and without clustering is comparatively high, ranging from 3.985s to 6.651s and from 4.993s to 7.592s, respectively. The results indicate that the 'Proposed Optimized QLRA-with clustering' consumes less delay over multiple iterations. The clustering technique proved to be effective in reducing the overall delay of the method.

Fig. 21 presents a comparative analysis based on Total Residual Energy for proposed and existing methods. Total Residual Energy is a critical metric that reflects the remaining energy in all network nodes, providing insights into the overall energy sustainability. Notably, the "Proposed Optimized QLRA-with clustering" maintains a higher Total Residual Energy, ranging from 50 J to 64J across nodes 50 to 250. In contrast, the "Optimized QLRA - without clustering" method achieves Total Residual Energy values ranging from 29 J to 48J. In comparison, both QLRA methods, with and without clustering, exhibit lower Total Residual Energy levels, ranging from 31 J to 16J and 21J to 7J, respectively. These results underscore the "Proposed Optimized QLRA-with clustering's" ability to preserve more significant Total Residual Energy compared to the existing methods, indicating its superior energy sustainability.

Complexity analysis

Fig. 22 presents the memory usage per node across varying node densities for different routing strategies. The proposed Optimized QLRA demonstrates the most efficient memory performance among all methods. For instance, at 300 nodes, the proposed method requires only 170 KB of memory per node, while QLRA without clustering consumes over 290 KB. This notable reduction is due to the efficient clustering mechanism and targeted Q-table maintenance, which avoids redundant neighbor and route storage. In contrast, the non-clustered variants experience higher memory consumption due to larger neighbor sets and the absence of localized aggregation. The results confirm that the proposed approach achieves better scalability and is well-suited for resource-constrained WSN environments.

4.1 Number of dead nodes and alive nodes against rounds

From Fig. 23 above, in comparison to existing routing methods, the "Proposed Optimized QRLA-with clustering" consistently outperforms them in maintaining a higher number of alive nodes and routing efficiency. Specifically, after 100 rounds, the proposed method retains 37 alive nodes, and 163 dead nodes surpassing the existing methods which only manage 35, 36, and 33 alive nodes and 165,164 and 167 dead nodes for "Optimized QRLA - without clustering," "QRLA - with clustering," and "QRLA - without clustering," respectively. This performance continues as the number of rounds progresses to 150, 200, 250, and 300. The "Proposed Optimized QRLA

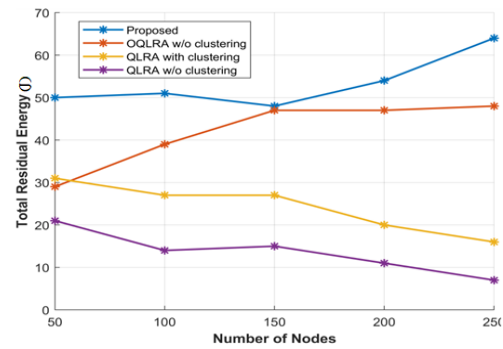


Figure. 21 Comparative Analysis based on Total Residual Energy

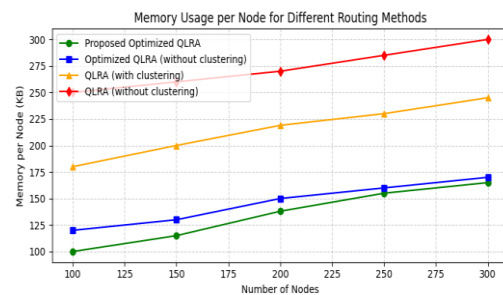


Figure. 22 Memory usage per node comparison for different routing methods

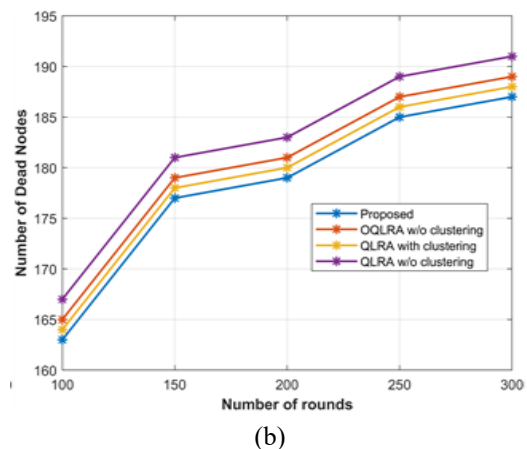
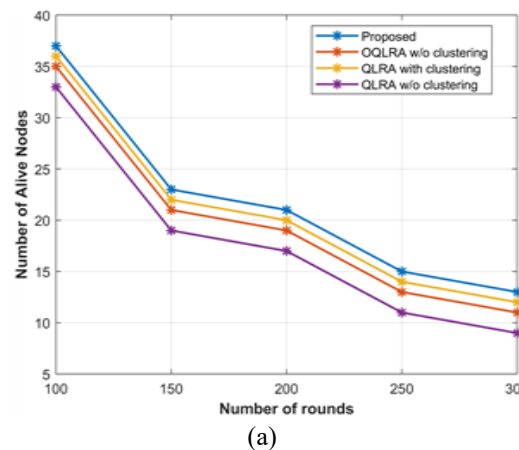


Figure. 23: (a) Number of Alive Nodes and (b) Number of Dead Nodes

Table 4. Comparative analysis of proposed work against state-of-the-art methods

References	Year	Throughput (Kbps)	Delay (s)	Network Lifetime (s)	Delivery ratio (%)	Total Residual Energy(J)	Energy consumption (J)
Sadek et al [16]	2021	23	-	2.6	-	56	-
Sureshand and Shyama [17]	2023	225	0.52	6.55	93.75	-	-
Dawood et al [20]	2021	-	1.6	-	83	65	-
Bairagi et al [24]	2024	63	4	-	73	-	-
Dogra et al [26]	2023			6		-	40
Thangavelu and Rajendran [27]	2024	-	2.35	5.107	98.05	-	80
proposed	2025	69	1.058	7.593	99.8	64	36

Table 5. Statistical performance (mean $\pm$ standard deviation) of different routing methods

Metric	Proposed Optimized QLRA	Optimized QLRA (without clustering)	QLRA (with clustering)	QLRA (without clustering)
Delivery Ratio (%)	0.924 $\pm$ 0.057	0.901 $\pm$ 0.061	0.832 $\pm$ 0.109	0.820 $\pm$ 0.062
Throughput (Kbps)	62.80 $\pm$ 5.85	56.80 $\pm$ 5.42	50.80 $\pm$ 7.47	45.60 $\pm$ 4.18
Network Lifetime (s)	6.930 $\pm$ 1.05	5.864 $\pm$ 0.94	3.653 $\pm$ 1.81	3.131 $\pm$ 0.98
Energy Consumption (J)	52.80 $\pm$ 5.04	61.20 $\pm$ 8.66	81.40 $\pm$ 4.80	93.40 $\pm$ 5.16
End-to-End Delay (s)	2.927 $\pm$ 1.13	3.846 $\pm$ 1.64	5.859 $\pm$ 0.94	6.923 $\pm$ 1.05
Total Residual Energy (J)	45.20 $\pm$ 7.63	36.60 $\pm$ 4.80	18.00 $\pm$ 5.59	4.80 $\pm$ 5.23

- with clustering" consistently maintains more alive nodes and fewer dead nodes in these scenarios, demonstrating its effectiveness in preserving network connectivity and vitality. The better performance of the "Proposed Optimized QRLA-with clustering" can be attributed to the utilization of clustering and optimization algorithms, making our proposed method a promising and more efficient choice for routing in various network scenarios when compared to the existing methods.

Scenario 4: Comparative Analysis with the state-of-art-methods

Table 4 shows the comparative analysis of our proposed work against state-of-the-art methods in terms of Delay, Delivery Ratio, Network Life Time, Throughput, Energy Consumption, and Total Residual Energy.

The proposed method demonstrates superior performance across multiple evaluation metrics compared to existing techniques, as summarized in Table 4. It achieves a throughput of 69 Kbps, which is higher than that of Sadek et al. [16], who achieved 23 Kbps, and Bairagi et al. [24], who achieved 63 Kbps, though slightly lower than the 225 Kbps achieved by Sureshand and Shyama[17]. In terms of delay, the proposed method records 1.058 seconds,

offering better performance than Thangavelu and Rajendran[27], who reported 2.35 seconds, and Bairagi et al. [24], who recorded 4 seconds, while being marginally higher than the 0.52 seconds recorded by [17]. For network lifetime, the proposed method achieves 7.593 seconds, outperforming [17] with 6.55 seconds and Thangavelu and Rajendran [27] with 5.107 seconds. The delivery ratio reaches 99.8 percent, the highest among all compared methods. Additionally, the total residual energy retained by the proposed method is 64 joules, close to the 56 joules retained in Sadek et al.[16], and significantly higher than the 40 joules in Dogra et al.[26]. These results clearly indicate that the proposed method offers balanced and consistent improvements in throughput, delay, network longevity, delivery reliability, and energy efficiency.

Scenario 5: Comparative Performance Analysis with Statistical Significance

Table 5 presents the statistical performance of the proposed Optimized QLRA in comparison with three other routing protocols: Optimized QLRA without clustering, QLRA with clustering, and QLRA without clustering. The metrics reported include the **mean** and **standard deviation** (SD) computed over

five independent simulation runs for each configuration.

The proposed Optimized QLRA consistently outperforms the other methods across all evaluation metrics. It achieves the highest packet delivery ratio of 0.924 ± 0.057 , indicating reliable data transmission even in dynamic conditions. It also records the highest throughput of 62.80 ± 5.85 Kbps, showing its effectiveness in maintaining high data flow. The high standard deviations across all metrics further indicate that the performance of the proposed method is stable and consistent across multiple runs, validating its robustness and reliability under varying network conditions.

Fig. 24 shows the convergence of ACOA based on best fitness value over iterations. The fitness improves rapidly in the early stages and gradually stabilizes around iteration 90, indicating that ACOA efficiently finds near-optimal solutions within 100 iterations. This behavior confirms that ACOA reliably converges toward high-quality solutions within a reasonable number of iterations, making it a suitable optimizer for action sequence selection in the proposed Q-learning-based routing framework.

5. Conclusion

This study proposed an innovative method that integrated clustering with an optimized Q-learning-based routing algorithm to enhance the energy efficiency of heterogeneous WSNs. Initially, CHs were selected using an attribute-based method, forming clusters that aggregate data and transmit it to the sink via Q-learning-optimized routes. To mitigate the issue of isolated nodes and enhance routing stability, the ACOA optimized the selection of forwarder nodes and action sequences within the Q-learning framework. The experimental evaluation, conducted under three simulation scenarios such as round variation, node density variation, and CH selection limits, demonstrated that the proposed method significantly outperforms existing techniques. The performance evaluation demonstrated that the proposed approach achieved higher delivery ratios, extended network lifetime, increased throughput, reduced delay, and improved energy efficiency, proving its effectiveness in enhancing overall network performance. Notably, it achieved a 25.19% reduction in energy consumption and an 11.82% increase in throughput compared to traditional and non-clustered QLRA methods. Additionally, the proposed method maintained a high delivery ratio up to 99.8% and preserved higher residual energy across all scenarios. The integration of ACOA allowed for

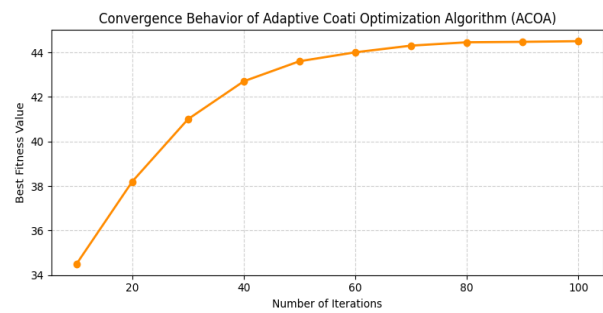


Figure. 24 Illustrates the convergence behavior of the ACOA

optimal forwarder node selection, which minimized routing delay and communication overhead, while the attribute-based clustering mechanism ensured balanced energy utilization. By leveraging ACOA to intelligently guide action sequence selection in Q-learning and integrating energy- and topology-aware cluster head selection, the method ensures more reliable routing paths, reduces communication overhead, and extends overall network longevity. The effectiveness and robustness demonstrated through simulation suggest strong potential for deployment in energy-sensitive WSN applications where stable and long-lasting communication is critical.

While the results demonstrated improvements in delivery ratio, network lifetime, throughput, delay, and energy efficiency, some limitations were identified. The reliance on a fixed attribute-based cluster head selection may not adapt well to dynamic network conditions, potentially leading to suboptimal performance in varying environments. Additionally, our approach does not currently account for fault tolerance, which could compromise network reliability in the presence of node failures. Future work could focus on integrating fault tolerance mechanisms to improve network reliability, implementing adaptive learning strategies to enable the routing algorithm to respond to dynamic conditions, and developing dynamic cluster head selection criteria based on real-time energy levels.

Conflicts of Interest

Authors must declare any conflicts of interest or explicitly state, "The authors declare no conflict of interest." Any personal circumstances or interests that could be perceived as having an undue influence on the presentation or interpretation of the research findings must be clearly disclosed.

Author Contributions

For research articles with multiple authors, a brief paragraph specifying the individual contributions of

each author must be included. The following format should be used: “Conceptualization, XXX and YYY; methodology, XXX; software, XXX; validation, XXX, YYY, and ZZZ; formal analysis, XXX; investigation, XXX; resources, XXX; data curation, XXX; writing—original draft preparation, XXX; writing—review and editing, XXX; visualization, XXX; supervision, XXX; project administration, XXX; funding acquisition, YYY,” etc. Authorship should be restricted to those who have made a substantial contribution to the work presented.

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